Human-AI Teaming Framework

As enterprise operations transition into the era of agentic AI and recursive learning, the primary risk to mission success shifts from mechanical failure to "Super-Huberous" autonomy—a state where AI models exhibit high confidence in probabilistic outputs that lack grounding in physical or logical reality. This document outlines a multi-disciplinary framework designed to move organizations from "Blind Reliance" to "Engineered Trust."

The strategy is anchored by three primary pillars:

1. **Trustworthiness Engineering:** A formal metrology treating trust as a measurable physical property, utilizing "Ground Truth Circuit Breakers" to veto AI intent that deviates from sensor-derived reality.
2. **Semantic Governance:** The industrialization of knowledge into governable assets through ontologies and knowledge graphs, providing "Semantic Guardrails" that force AI to align with physical constants and business rules.
3. **Human-AI Teaming Framework:** A supervisory system that monitors human performance to prevent the "Automation Paradox," where improved AI accuracy leads to human complacency, skill atrophy, and a subsequent decline in overall system reliability.

Financial analysis suggests that implementing these safeguards prevents substantial costs—estimated at $10–20 million over three years—by mitigating risks associated with model collapse, alert fatigue, and catastrophic system failures.

The Trustworthiness Engineering Framework

Trustworthiness Engineering applies metrology to non-deterministic digital assets to quantify system reliability. It shifts the engineering focus from "Performance-First" to "Observability-First."

The AI Circuit Breaker (Deterministic Veto) - *covered in detail in other docs*

The core of this architecture is a hard-coded safety layer that acts as a final gatekeeper for AI actions.

- **Logic Gate:** The system evaluates the discrepancy (**Delta**) between AI **Intent** (strategic/tactical planning) and **Sensor State** (ground truth).
- **Veto Logic:** `IF (Delta > Tolerance) OR (Trust Metric < Minimum Threshold) THEN ACTIVATE CIRCUIT BREAKER`.

- **Deterministic Authority:** In the event of a breach, sensors are granted absolute authority to override probabilistic AI suggestions, initiating a "System Halt" or activating a "Safe State."

Metrology and Key Metrics

Trust is measured through specific quantitative data points:

- **Mean Time Between Hallucinations (MTBH):** The average operating time before a system produces a high-confidence error.
- **Value-Drift Coefficient:** A numerical measure of the distance between an autonomous agent's current objective and the original human intent.
- **Intent Delta:** The measured difference between AI logic and physical reality.
- **The Rule of Ten:** Sensory precision must be an order of magnitude higher than operational tolerance.

Semantic Governance and Knowledge Engineering

Semantic Governance establishes a machine-readable language for business concepts, harmonizing ontologies and taxonomies across the enterprise.

Semantic Guardrails

By using Knowledge Graphs as the "Physical Laws" of logic, the system ensures AI remains grounded.

- **Beyond Probability:** While LLMs operate on what word is *likely* to follow, semantic guardrails define what *must* be true based on physics and engineering.
- **Five Dimensions of Relevancy:** To achieve contextual relevance, the architecture captures:
    1. **Who:** Persona Model (User role).
    2. **What:** Business domains and sub-domains.
    3. **When:** Time model (past, present, and future intent).
    4. **Where:** Location model.
    5. **Why:** Intent/Causality model (Reasoning chains).

Architectural Components

Layer	Description	Key Technologies
Knowledge Fabric	Foundational semantic modeling and storage.	Metaphactory, GraphDB

Operational Workbench	Low-code platform for monitoring and interaction.	Tom Sawyer Perspectives
Intelligence Layer	Supports semantic search and AI orchestration.	Vector DBs, AI Agents
Integration Backbone	Secure communication between microservices.	Model Context Protocol (MCP)

Human-AI Teaming: Mitigating the Automation Paradox

A critical failure in automated systems is the "Automation Paradox": as AI accuracy improves (e.g., from 65% to 92%), human supervisors become complacent, their manual skills atrophy, and system reliability eventually drops (e.g., from 95% down to 85%) because humans fail to catch the remaining 8% of errors.

Human Performance Monitoring

To prevent supervisory failure, the framework instruments human performance in real-time:

- **Fatigue Scoring:** A weighted sum (0–1 scale) where scores >0.7 trigger mandatory 15-minute breaks.
    - **Session Duration (30%):** Maxed at six hours.
    - **Time Since Last Break (30%):** Maxed at 90 minutes.
    - **Accuracy Degradation (20%):** Drops in baseline performance.
    - **Decision Speed Increase (20%):** Indicators of "rushing."
- **Trust Calibration:** The system monitors if an engineer's acceptance rate of AI suggestions matches the AI's actual accuracy. If an engineer accepts 94% of suggestions in a range where the AI is only 71% accurate, the system issues a context-specific over-trust warning.

Competency Maintenance

To prevent skill degradation, the framework requires:

- **Manual Exercises:** Every 14 days, engineers must perform manual mapping without AI assistance.
- **Evaluation:** Exercises are scored on correctness (60%) and reasoning quality (40%).
- **Remediation:** If scores fall >15% below baseline, training is automatically scheduled.

Operational Implementation: CloudOps Pro Managed Infrastructure

The application of these principles is reflected in high-stakes managed services contracts, specifically the "CloudOps Pro LLC" framework for Enterprise Corporation.

Contractual SLAs and Performance Credits

The contract (VNDR-2026-0042) defines strict availability and response targets:

- **Infrastructure Availability:** 99.95% monthly uptime.
    - 99.90% - 99.94% availability: 5% fee credit.
    - 99.80% - 99.89% availability: 10% fee credit.
    - < 99.80% availability: 20% fee credit + Performance Improvement Plan (PIP).
- **Incident Response:** P1 incidents must be acknowledged within 15 minutes (measured by median).
    - 16–30 minutes: Warning (3 strikes leads to performance review).
- **Cost Optimization:** 15% reduction achieved by month 6. Failure to reach at least 10% requires contract renegotiation.

Observability Automation

The contract requires the use of Terraform modules (vendor_observability_space) to automatically provision dashboards and Service Level Management integrations. These monitors trigger alerts for SLA breaches, documenting incidents in vendor scorecards and initiating escalation protocols via Slack and PagerDuty.

Risk Mitigation and Recursive Integrity

Preventing Model Collapse

Error in AI systems is not static; it propagates and amplifies. To prevent "Model Collapse"—where an AI trained on its own hallucinations loses the ability to distinguish reality from noise—the framework employs **Recursive Integrity**:

- **Data Exclusion Filter:** Any data point associated with a "Circuit Breaker Trip" is purged from the training queue.
- **Recursive Loop Sanitization:** Only "Clean Data" (validated ground-truth successes) is permitted to update the agent's internal logic.

Organizational Safeguards

Management dashboards track "Production Pressure" to identify systemic risks.

- **Pressure Indicators:** High decision speeds, accuracy declines, and excessive alert overrides.
- **Interventions:** If team pressure scores exceed 0.7, non-critical work is halted, and daily targets are reduced by 30–50%.

Implementation Timeline and ROI

The framework requires an eight-month implementation phase with a total investment of approximately $1.2 million.

- **Preventable Costs:** $10–20 million (remediation, rework, and prevention of catastrophic failures).
- **ROI:** 8x to 17x over three years.
- **Intangible Benefits:** Regulatory compliance evidence, reduced engineer burnout, and a scalable foundation for future AI initiatives.